Meta Hackthon 2026

Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System

Meta Muse Vision AI — Wearable AI-Powered Visual Intelligence System

Voice-Powered Visual Intelligence | 3D Print Wearable Camera + Android + Multimodal AI

Detailed Project Concept, Physical Prototype, Technologies, Real-World Applications, Usability, and Future Engineering Roadmap

1. Overview

Even in the AI era, many industries **still rely on traditional hardware barcode scanners** with limited capabilities. **Muse Vision AI** introduces a wearable, **voice-powered smart headset** that goes beyond **barcode scanning to recognize objects**, **count items**, and **interpret visual information**. Combining a custom 3D-printed headset, Android application, and Meta Muse AI, the solution **enables hands-free intelligent scanning across multiple industries**. It aims to transform conventional scanning into a flexible, AI-powered visual intelligence experience.

2. Problem Statement

Industries continue to rely on **expensive dedicated barcode scanners** and **wearable scanning devices** for **simple identification tasks**. **Commercial handheld and mobile scanners** can cost over **$100–$3000**, while some **industrial wearable scanners** cost **$800–$1,300** or more.

Wireless wearable devices may also require battery charging, Bluetooth pairing, connection troubleshooting, and ongoing maintenance. Losing or damaging these devices increases replacement costs and operational downtime.

Traditional **barcode scanners primarily decode barcode values** and **often require additional software or API lookups to understand associated information**. Users need a **low-cost, plug-and-play solution** that eliminates **unnecessary hardware complexity** while **supporting multiple barcode scans, associated label information, and intelligent visual tasks.**

3. Proposed Solution — Meta Muse Vision AI

Meta Muse Vision AI introduces an affordable, AI-powered wearable scanning system combining a custom 3D-printed headset, USB camera, Android application, and Meta Muse AI.

The hardware prototype is designed to cost approximately $45–$50, with the following estimated components:

- **3D-printed headset: $10**
- **USB camera: $30**
- **USB cable: $5**
- **Estimated total: $45**

The **camera connects directly to an existing Android phone** through USB-C**, eliminating the need for a separate headset battery, charging system, Bluetooth pairing, Blurry live preview or wireless camera configuration.**

Users can simply plug in the camera and start scanning. The lightweight, replaceable device can be worn as a headset or mounted at a cashier counter, packing station, or delivery vehicle.

With Muse AI, the solution goes beyond **conventional barcode scanning** to **support multiple barcode recognition**, **extraction of associated printed identifiers**, **object counting**, and **visual information analysis**. The extracted information can also integrate with existing business applications and APIs.

Our vision is simple: Replace expensive, specialized scanning hardware with an affordable, replaceable, plug-and-play camera powered by intelligent software.

Note: The estimated hardware cost excludes the Android phone, software operation, and AI service costs. Cloud-based Muse AI processing still requires internet connectivity through the phone.

4. Components / Technologies

Hardware Components

- **USB8MP05AF 8MP** Autofocus UVC Camera (~$30): **Captures images and videos** for barcode recognition, object counting, and AI-based visual analysis.
- Custom 3D-Printed Headset (~$10): **Lightweight**, **wearable camera mounting frame** designed using Blender and manufactured using a 3D printer.
- USB Cable / USB-C Connector (~$5): **Connects the camera directly to an Android smartphone**, providing both power and data transfer.
- Android Smartphone: Runs the companion application, processes voice commands, provides GPS information, and communicates with Muse AI.

Software & Technologies

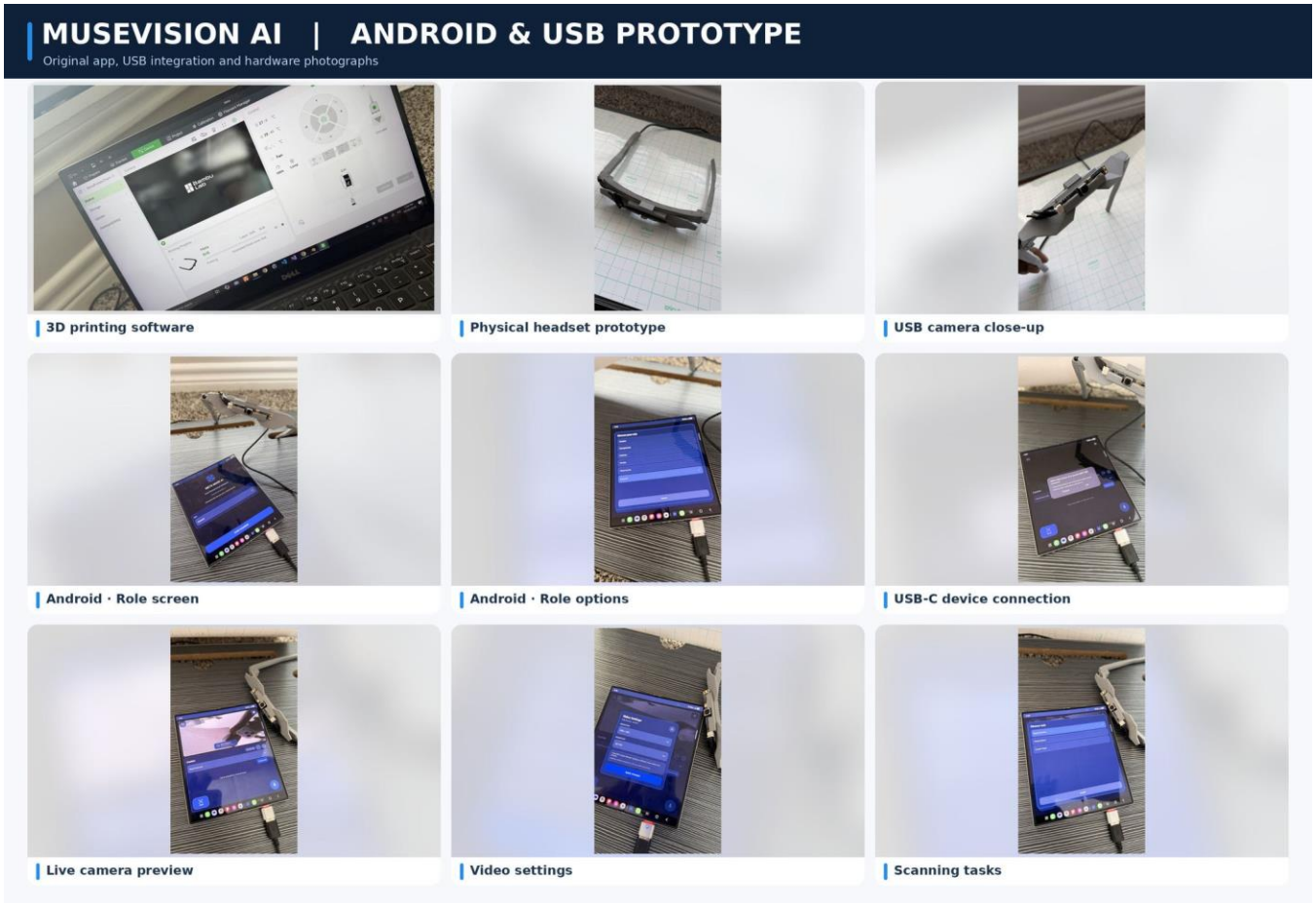
- **Meta Muse Spark AI**: Multimodal AI for **image understanding, object recognition, counting, and extracting barcode-associated information**.
- **.NET 10 MAUI / C#**: Cross-platform application framework used to develop the Android and iOS companion application, scanning workflows, and user interface.
- Android USB Host / UVC Integration: **Enables direct USB camera connectivity and live video preview** without Bluetooth or Wi-Fi pairing.
- Android Speech Recognition: **Enables hands-free scanning through voice commands.**
- Android GPS & Geolocation: **Supports delivery address verification against the smartphone's current location.**
- Blender: **Used for precise 3D modeling**, camera positioning, **headset dimensions**, foldable supports, and **printable mechanical components**.
- 3D Printing / PLA Filament: Used to manufacture and refine the custom headset prototype.

Meta Hackthon 2026

Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System

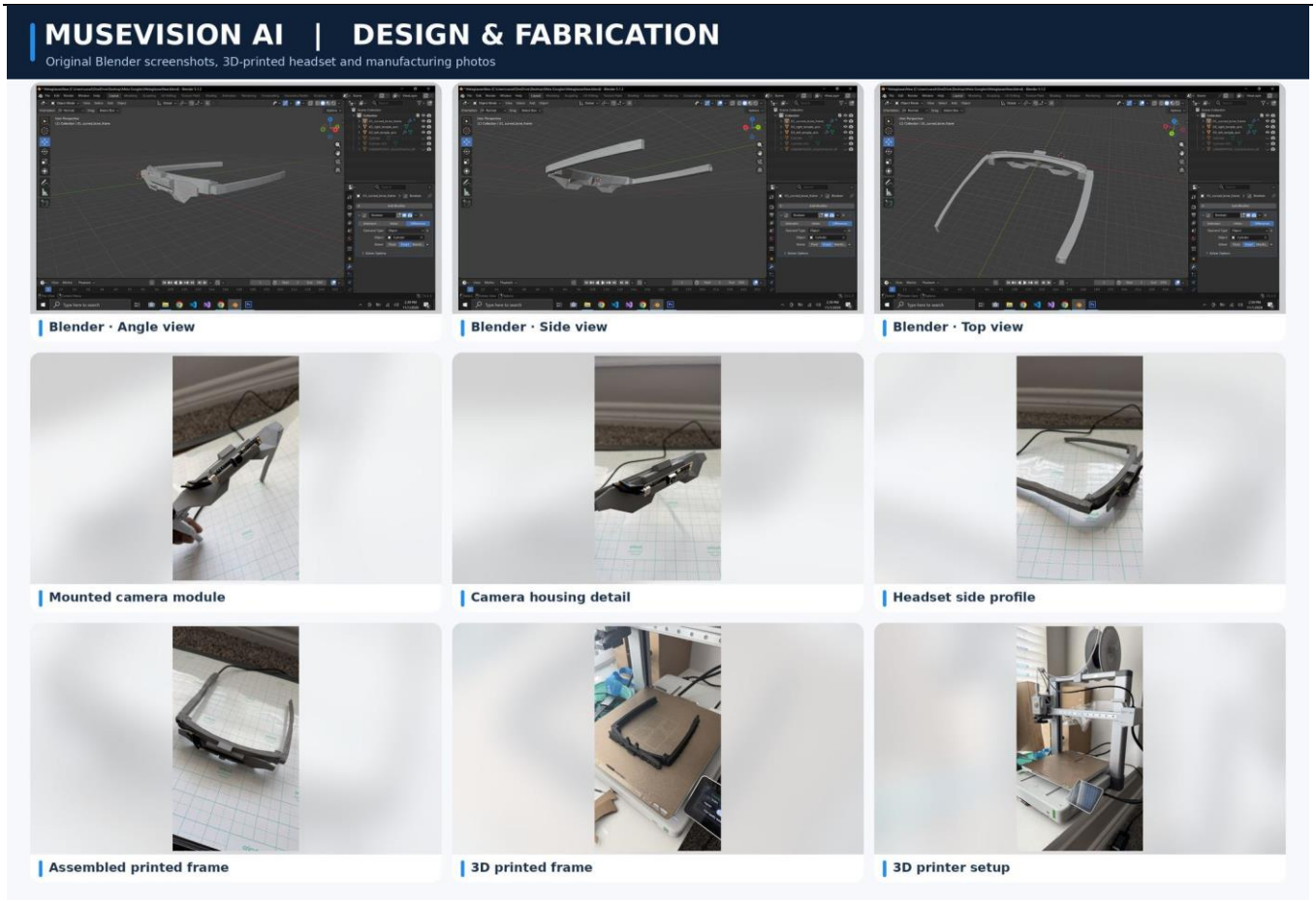
Prototype Development – 3D Models, Printed Hardware & Application Screenshots

The following images document the design, development, and physical implementation of the Meta Muse Vision AI wearable scanning system for Meta Hackathon.



Meta Hackthon 2026

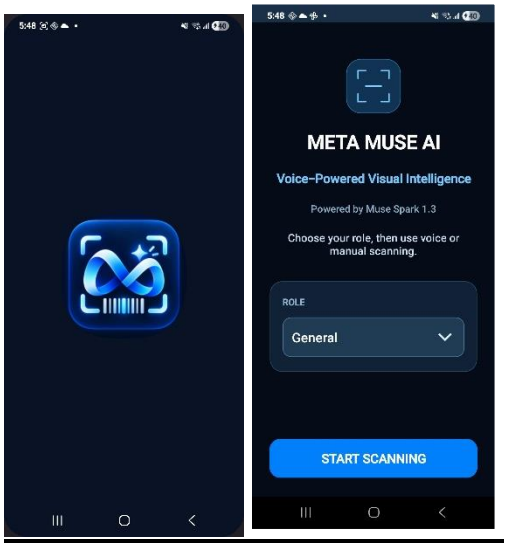
Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System



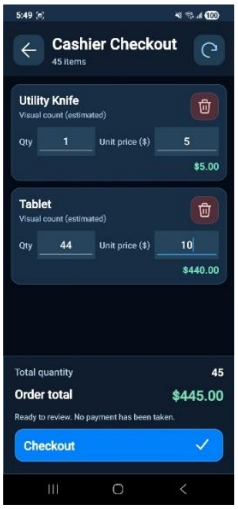
Meta Hackthon 2026

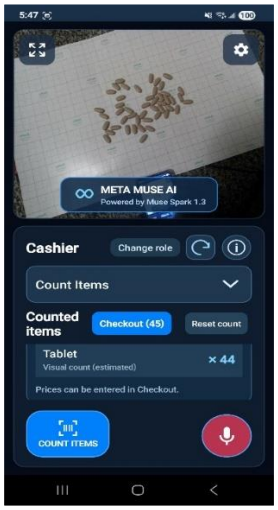
Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System

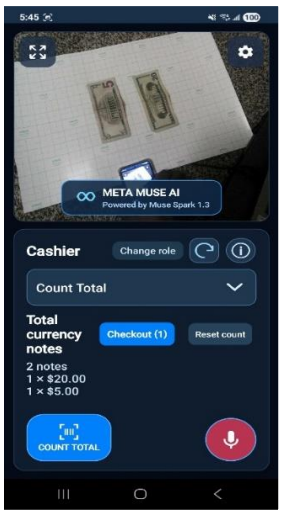
Mobile App Screenshots: Developed using .NET MAUI, Meta Muse AI is designed for cross-platform use, enabling its USB-connected smart headset to work with compatible smartphones, tablets, and computers.



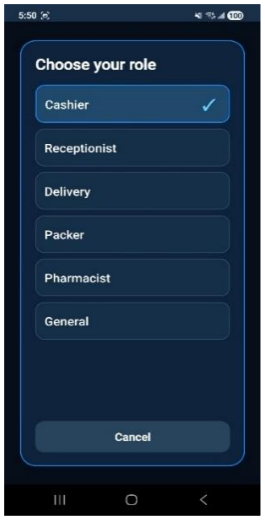












Meta Hackthon 2026

Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System

Real-world applications demonstrate how the headset can be used across different industries.



Meta Hackthon 2026

Muse Vision AI
Wearable AI Vision & Voice-Powered Scanning System

YouTube – Project Demonstration

Watch the Meta Muse Vision AI prototype and live demonstration.
https://youtu.be/AcdOGsPkFgk

GitHub – Project Source Code

Explore the .NET MAUI Android application, Meta AI integration, and complete project source code.
https://github.com/varadacharisuresh/MetaMuseVisionAI

LinkedIn – Developer Profile: Suresh Kumar Varadachari

Connect with the developer and explore professional experience, technical expertise, and innovative projects.
https://www.linkedin.com/in/suresh-kumar-varadachari-0a9b9129/